

Please make
Posit
account

- Go to <https://posit.cloud>
- Sign in (or create a free account)

Access
Github

- <https://github.com/Hanson-Lab/Microbiome-Visualization-Workshop>
- QR code available



Using R to Analyze and Display Microbiome Data – a Workshop

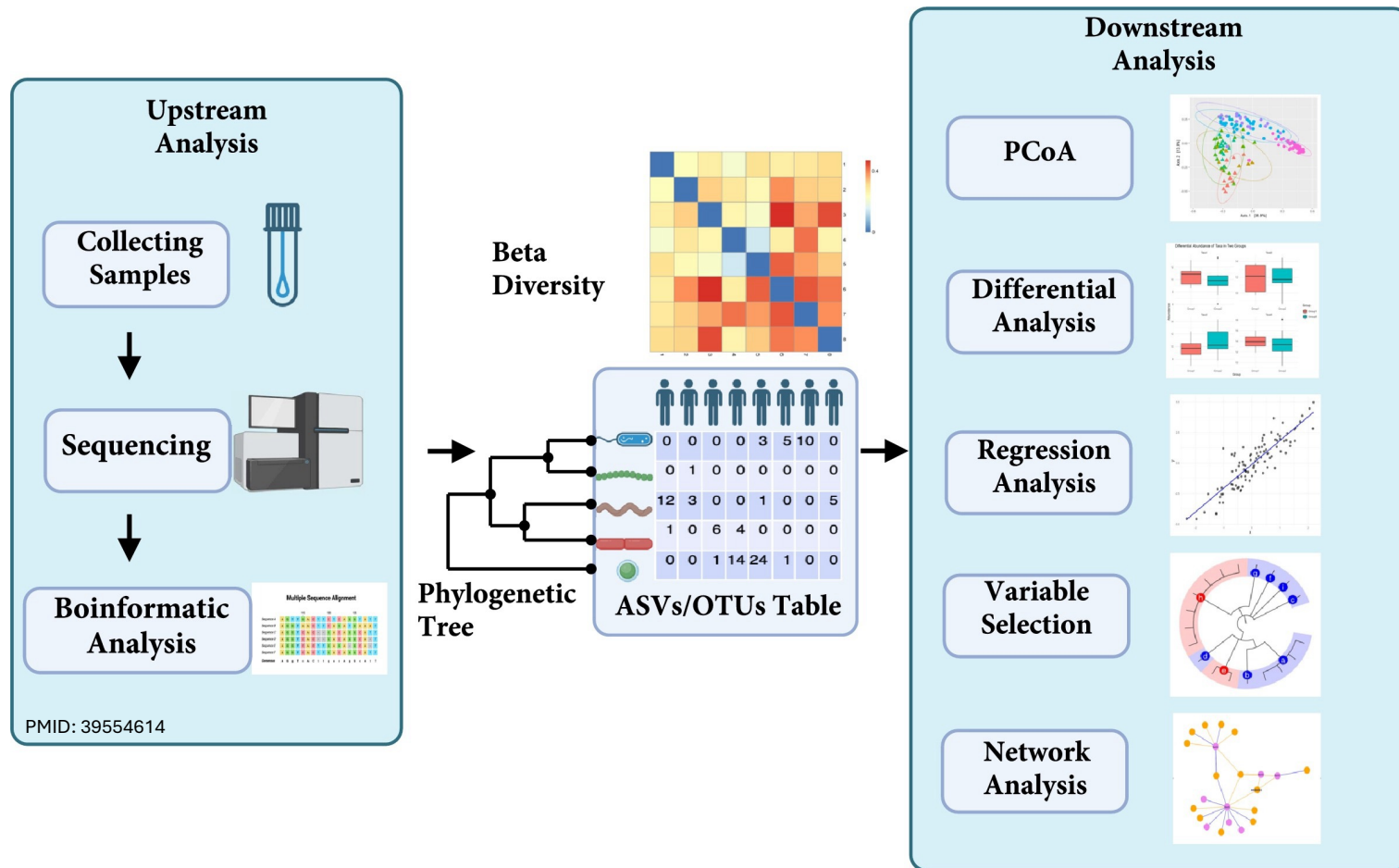
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 **UTHealth** | **School of Public Health**
The University of Texas
Health Science Center at Houston

**Gulf Coast
Consortia**
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SCIENCES

Overview



Brief overview of microbiome data structures

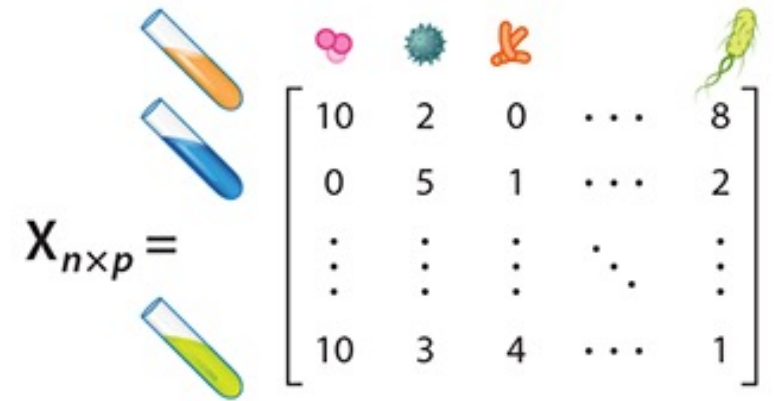
- Advancements in next-generation sequencing methods have expanded approaches used to analyze the human microbiota
- However, premise remains:
 - Bacteria in a sample are identified and quantified to obtain abundance values for taxonomy of interest (microbiome features, p) in each sample (n , patient stool samples, for example)
 - $n \times p$ matrix



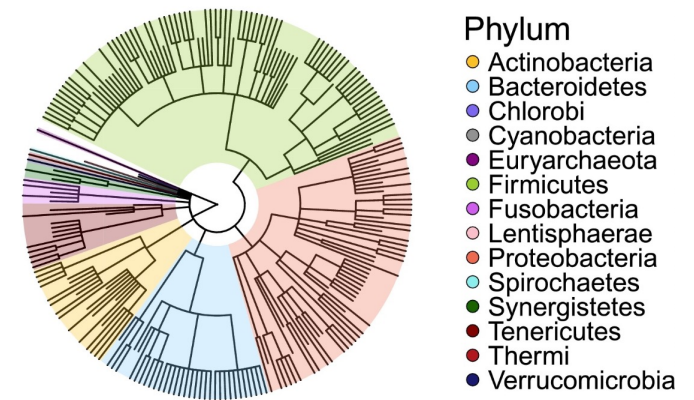
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Key components of microbiome data structures

- Count based
 - Dependent on library size and therefore represent relative counts
- Compositional
 - Fixed sum of bacterial counts in each sample
- Zero-inflated
 - Features in one sample may not appear in another
 - Rare features
- Highly dimensional
 - $p \gg n$; number of features greater than n samples
- Tree-structured
 - Can be structured into trees to provide meaningful information on taxonomy and evolutionary relationships



PMID: 38962089



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Activating R environment and loading data into R programming

- **Step 1 — Launch Posit Cloud**

- Go to <https://posit.cloud>
- Sign in (or create a free account)

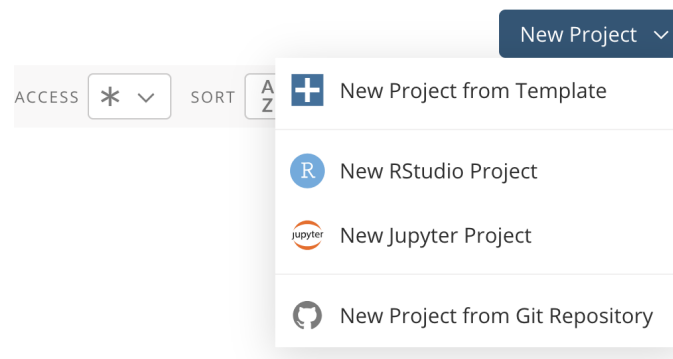
- **Step 2 — Create Pre-Meeting Workshop Posit Template**

- <https://tinyurl.com/AMR-Workshop>

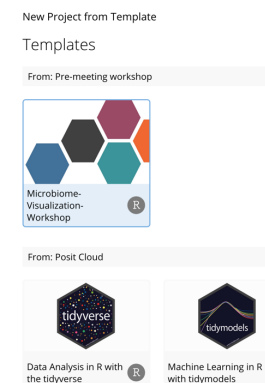
- **Step 3 – Activate Pre-Meeting Workshop Posit Template**

- 1) Click “New project from template” then 2) “Microbiome-Visualization Workshop”

1)



2)



Data pre-processing measures for microbiome analysis

- After quality control measures
 - Identifying microbial contaminants
 - Removing samples that were under sequenced
- Create phyloseq object
 - Need the following
 - `otu_table()` – raw read/amplicon counts (taxa x samples)
 - `meta_table()` – sample-level information used for grouping and comparisons
 - `taxa_table()` – taxonomic labels for each OTU



Describing the microbiome: Summarizing composition by taxonomy

- Diversity analysis
 - Relative abundance of taxonomy and functional pathways
 - Alpha diversity (within sample diversity)
 - Beta diversity (between sample compositional comparisons)

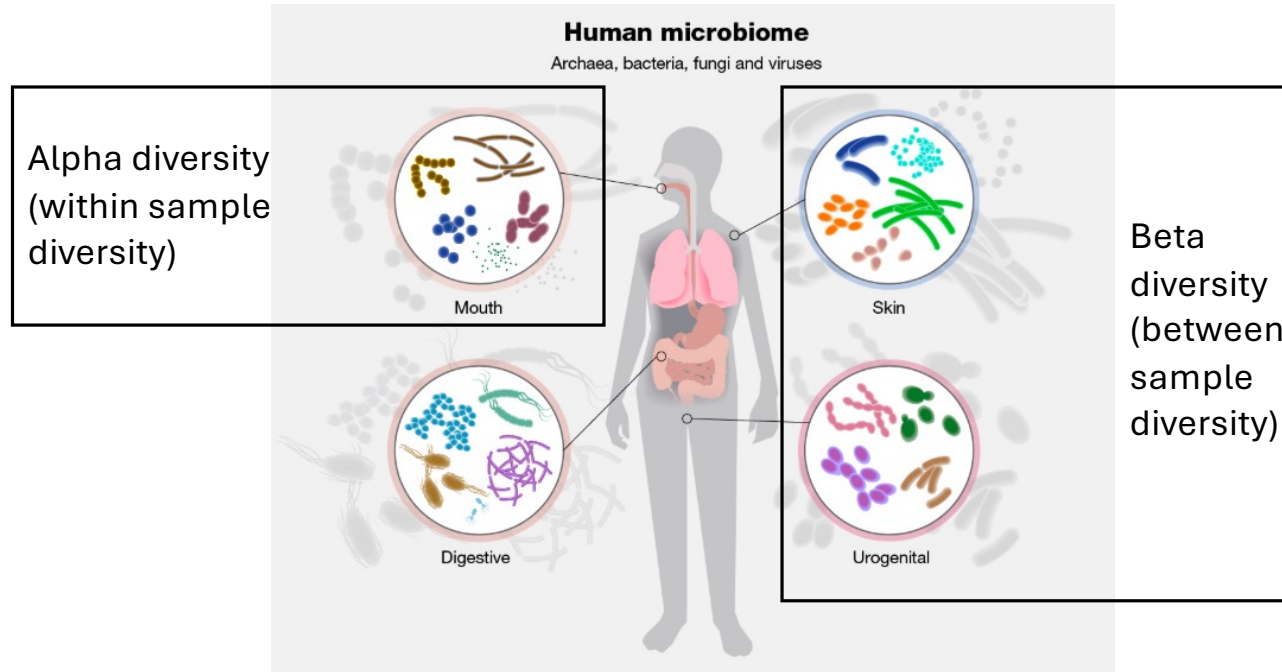
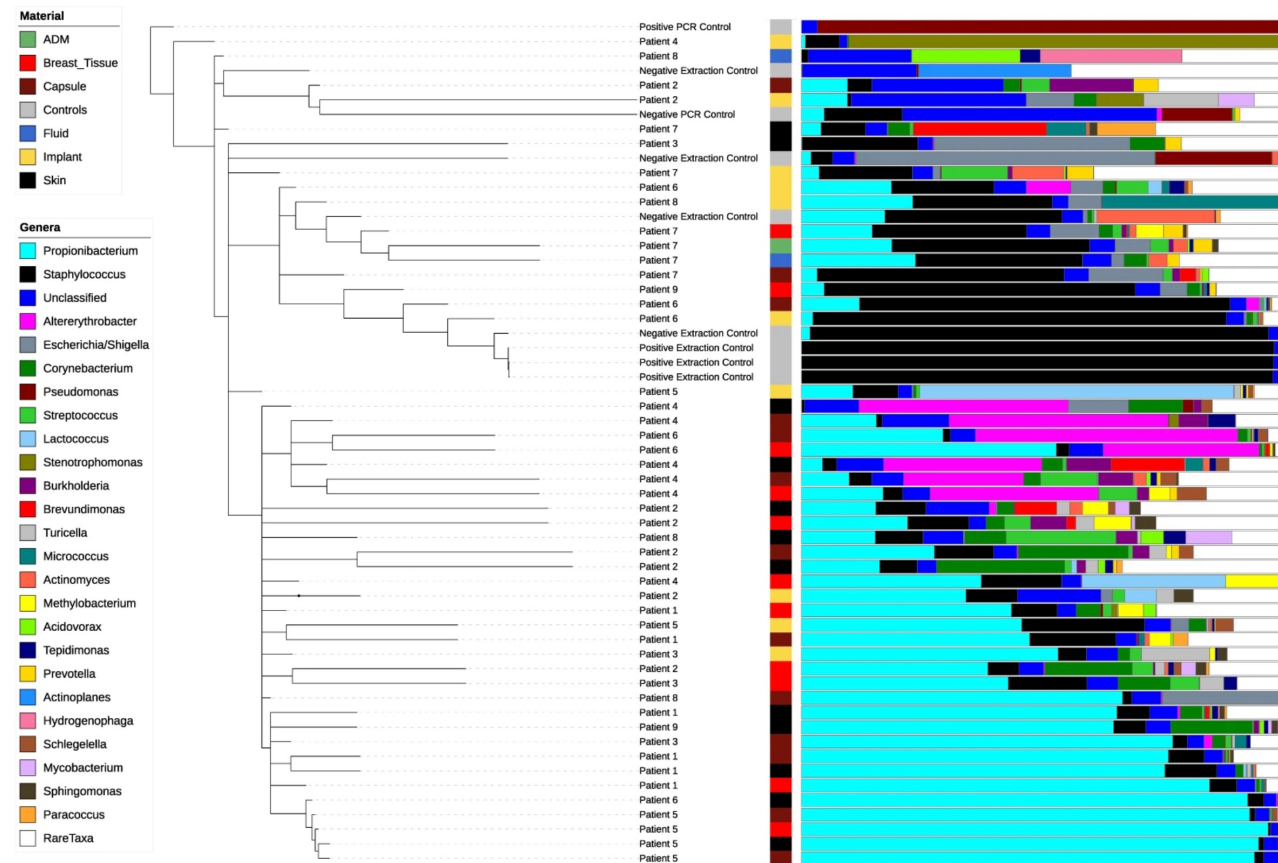


Image adapted from: <https://www.genome.gov/genetics-glossary/Microbiome> and
<https://bioinformatics.ccr.cancer.gov/docs/qiime2/Lesson5/>

Describing the microbiome: Visualizing community composition

- Stacked bar plots are a commonly used to visualize overall composition



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Diversity Analysis – Alpha diversity

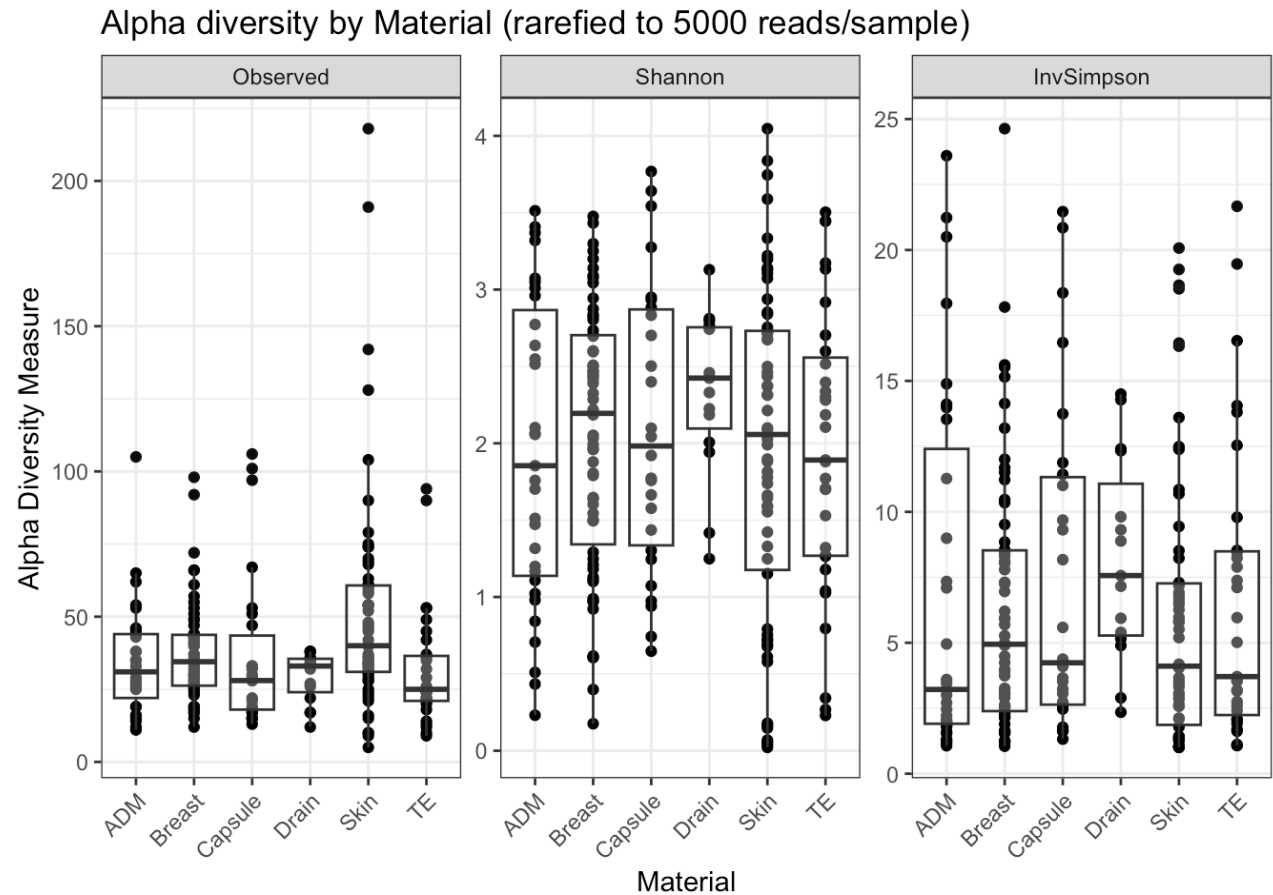
- Alpha diversity – within sample diversity
 - *To rarefy or not, that is the question?*
- Since alpha diversity is strongly dependent on sequencing depth we rarefy
 - This is only performed for alpha diversity comparisons, other downstream analysis do not require rarefaction
 - See McMurdie and Holmes paper
- Rarefaction involves subsampling reads without replacement to a defined sequencing depth to create a more standardized library size
 - Minimum total read depth across all samples
 - Samples with less than this value are discarded

Diversity Analysis – Alpha diversity

- After rarefaction we can now calculate alpha diversity metrics

- Multiple metrics

- Observed (richness)
- Shannon
- Inverse Simpson
- Simpson
- Chao1



Diversity Analysis – Alpha diversity mathematical equations

$$\textit{Shannon} = - \sum_{i=1}^R p_i \ln p_i$$

$$\textit{Simpson} = \sum_{i=1}^R p_i^2$$

$$\textit{Inverse Simpson} = 1 / \sum_{i=1}^R p_i^2$$

$$S_{chao1} = S_{obs} + \frac{N_1^2}{2N_2}$$

p_i = proportion of observations (reads, or abundances) belonging to the i th species

S_{obs} = the number of observed species

N_1 = the number of singletons (species captured once)

N_2 = the number of doubletons (species captured twice)

Between sample comparisons: Beta diversity

- Calculate dissimilarity distance metric (Bray-Curtis, Jaccard, UniFrac)
- Visualization or statistical test
 - ordination plots (PCoA or NMDS)
 - PERMANOVA
 - test of homogeneity

Beta Diversity: Distance metrics

- Available distance metrics include

- Bray-Curtis

- Examines abundance of microbes that are shared between samples and the number found in each sample
 - Ranges from 0 to 1

$$\text{Bray Curtis} = 1 - \frac{2C_{ij}}{S_i + S_j}$$

C_{ij} = sum of the lesser values for species in common between two samples
 S_i and S_j = total number of microbes found in samples i and j

- Jaccard

- Takes into consideration the presence of microbes in one or multiple samples
 - Ranges from 0 to 1

$$\text{Jaccard}_{AB} = 1 - \frac{|A \cap B|}{|A \cup B|} = 1 - \frac{|A \cap B|}{|A| + |B| - |A \cap B|}$$

- UniFrac

- Considers the phylogenetic relationship between microbes
 - Weighted and unweighted

- Bray-Curtis commonly used

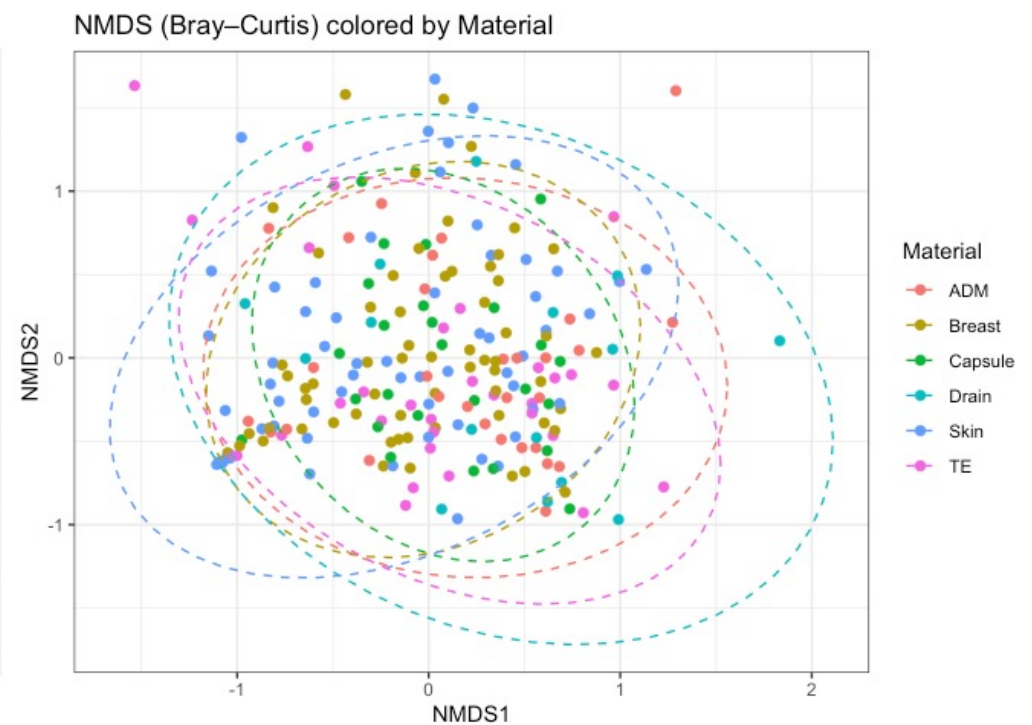
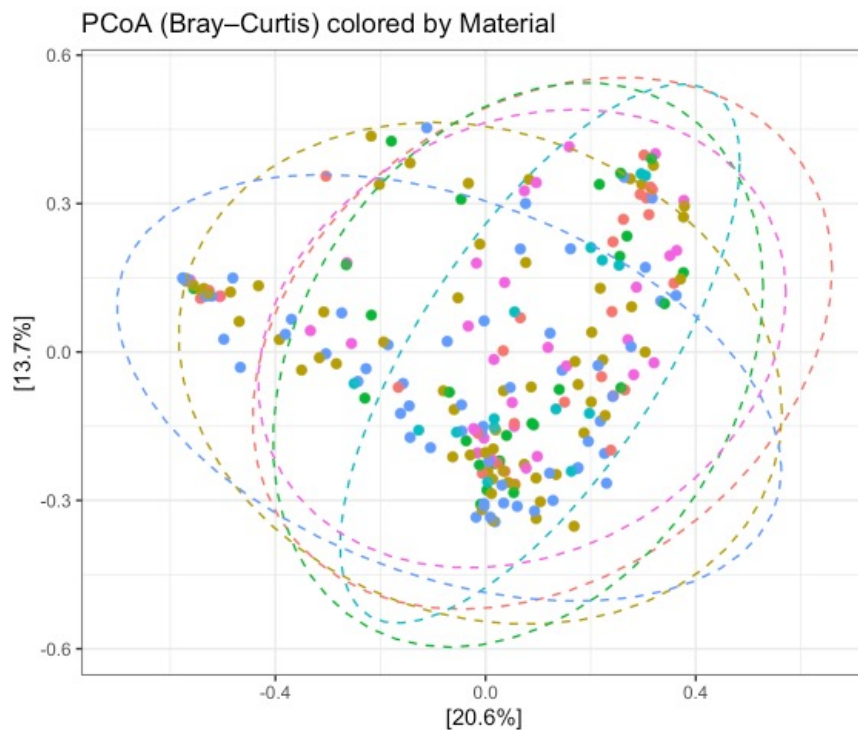
- Strongly influenced by library size so we use relative abundances to limit effect

Between sample comparisons: Ordination methods

- Ordination plots
 - Dimensionality reduction techniques
- PCA – Principal Component Analysis
 - Assumes linearity, identifying principal components that are linearly independent
- PCoA – Principal Coordinates Analysis
 - Useful for looking at the dissimilarity/similarity in sample composition
- NMDS – Non-Metric Multidimensional scaling
 - Rank based dimensionality reduction

Between sample comparisons: Ordination methods

- Here we illustrate the visualization of between sample variation using
 - Bray-Curtis distance metric combined with PCoA and NMDS



Detecting and visualizing microbiome associations

- Network analysis
- Popular for visualizing microbiome interactions within a community
 - Specifically, correlation or co-occurrence networks
- **What is co-occurrence?**
 - A statistical association between taxa across samples
 - Often summarized as a correlation (positive and negative)
 - A way to describe patterns of co-variation
- **What co occurrence is not?**
 - Proof that taxa interact directly
 - Evidence of competition or cooperation
 - Causal inference

Correlation methods for microbiome associations

- Commonly used methods:

- Pearson

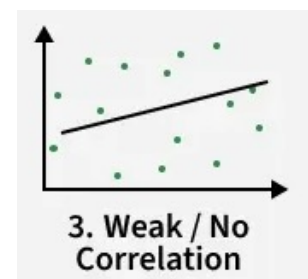
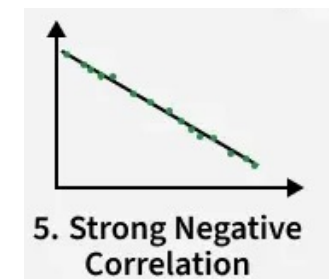
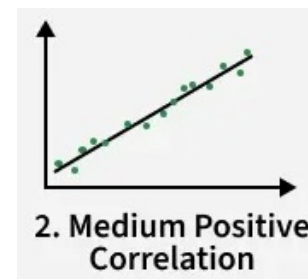
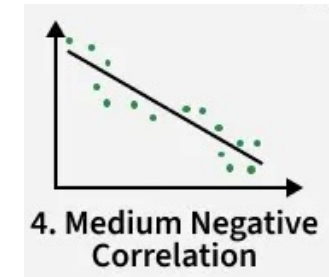
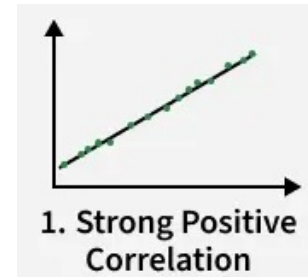
- Measures the relationship between continuous, normally distributed linearly related variables (parametric)

- Spearman

- Measures the strength and direction of a monotonic relationship using continuous or ordinal data (non-parametric)

- Kendall rank

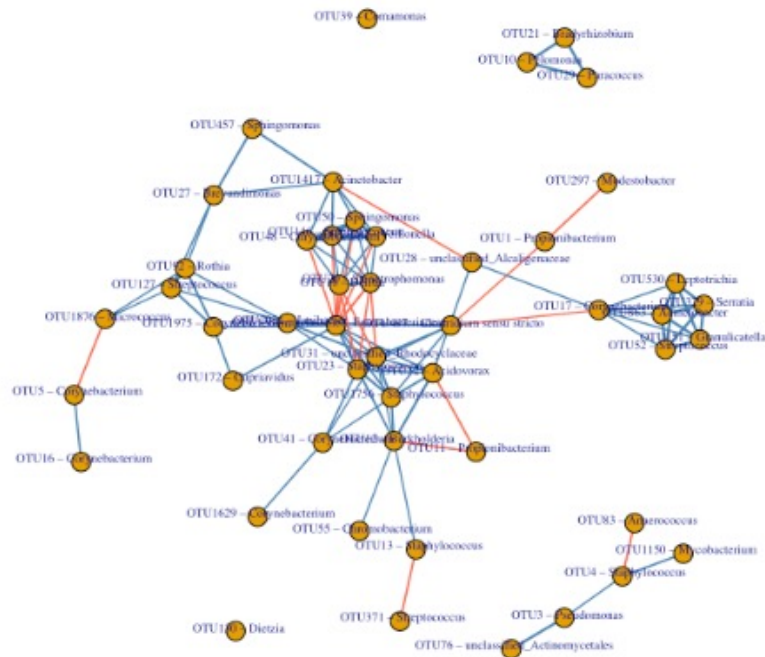
- Measures strength of dependence between two ordinal variables (non-parametric)



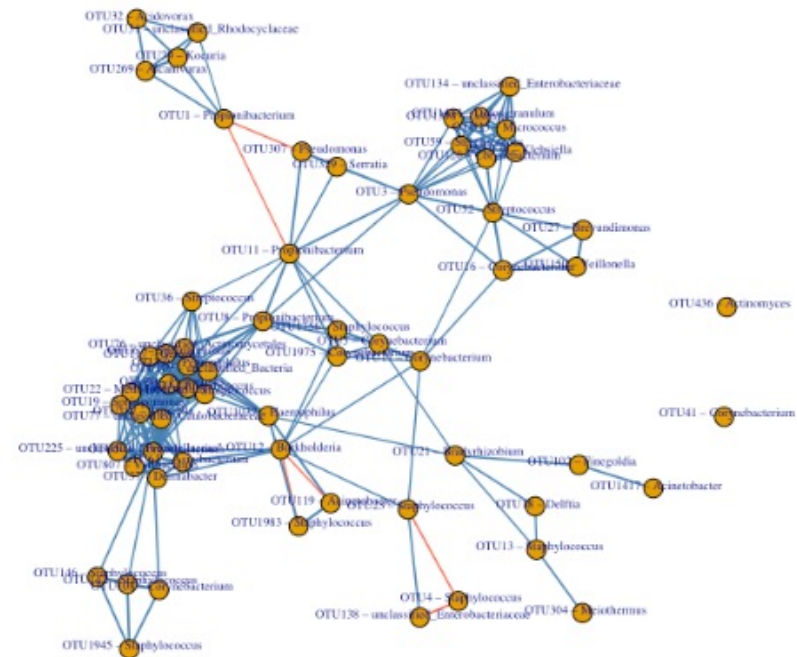
Visualizing microbiome associations

- Here we used spearman for the correlation analysis and built a correlation network using the igraph package

Co-occurrence network (ADM + No cancer): ABX = No



Co-occurrence network (ADM + No cancer): ABX = Yes



Interpreting co-occurrence networks

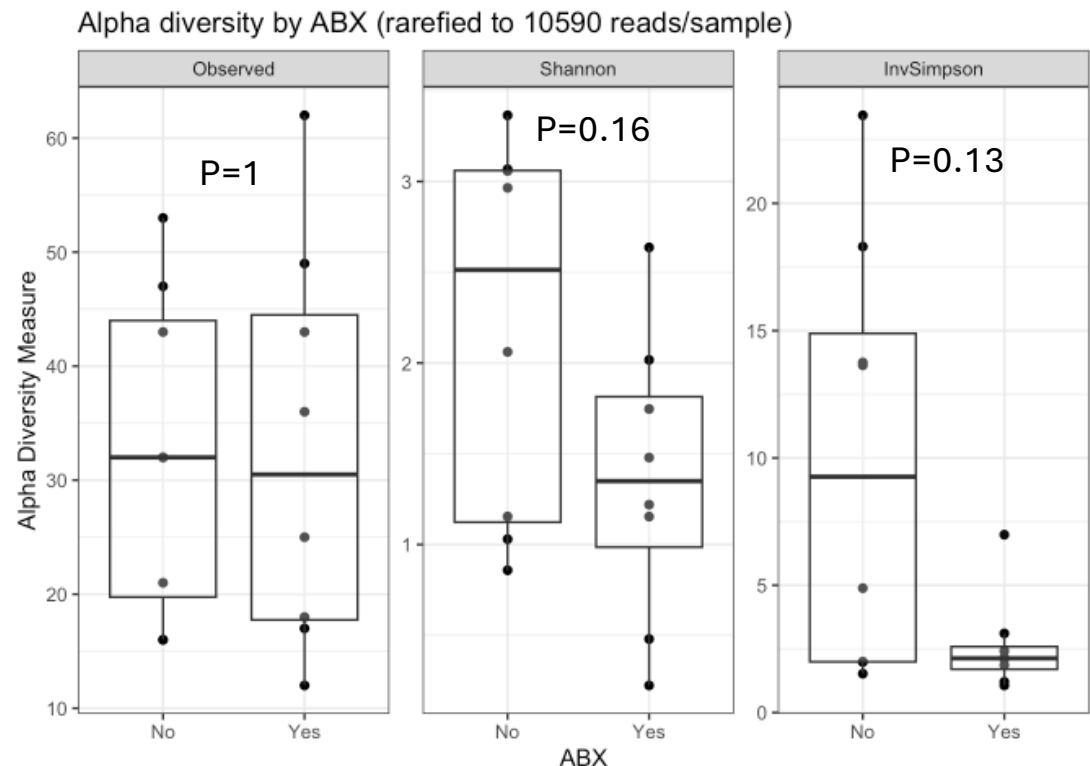
- Helpful to quantify the overall structure using simple summary metrics, like
 - Density
 - How connected a network is overall
 - Calculated as the proportion of observed edges relative to the maximum number of possible edges in the network
 - Density ranges from 0 to 1
 - Density ≈ 0 is a very sparse network (few taxon–taxon associations detected)
 - Density ≈ 1 is a very dense network (many taxa connected to many others)
 - Transitivity
 - Global clustering coefficient that measures the tendency of nodes to form closed triplets, or “triangles”
 - Like density, transitivity ranges from 0 to 1:
 - Low transitivity (≈ 0) indicates associations are mostly linear or chain-like
 - High transitivity (≈ 1) indicates taxa tend to form tightly connected clusters

##	nodes	edges	density	transitivity
## ABX = No	49	104	0.08843537	0.5760870
## ABX = Yes	62	236	0.12480169	0.7133676

Adding metadata to microbiome associations: Alpha diversity

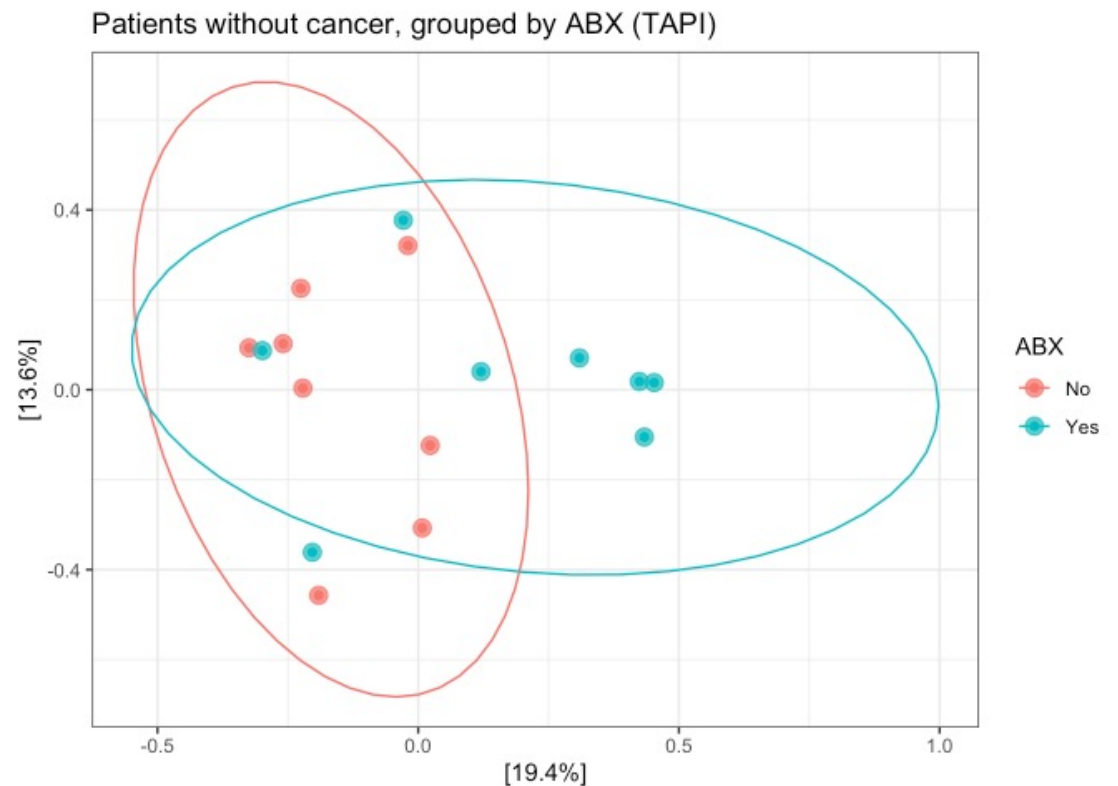
- We are interested in patients without cancer that were given Triple Antibiotic Pocket Irrigant (TAPI) or ABX
- Let us compare species diversity in each sample between our groups of interest. First:

- Rarefy
- Estimate richness
- Compare using Wilcox test



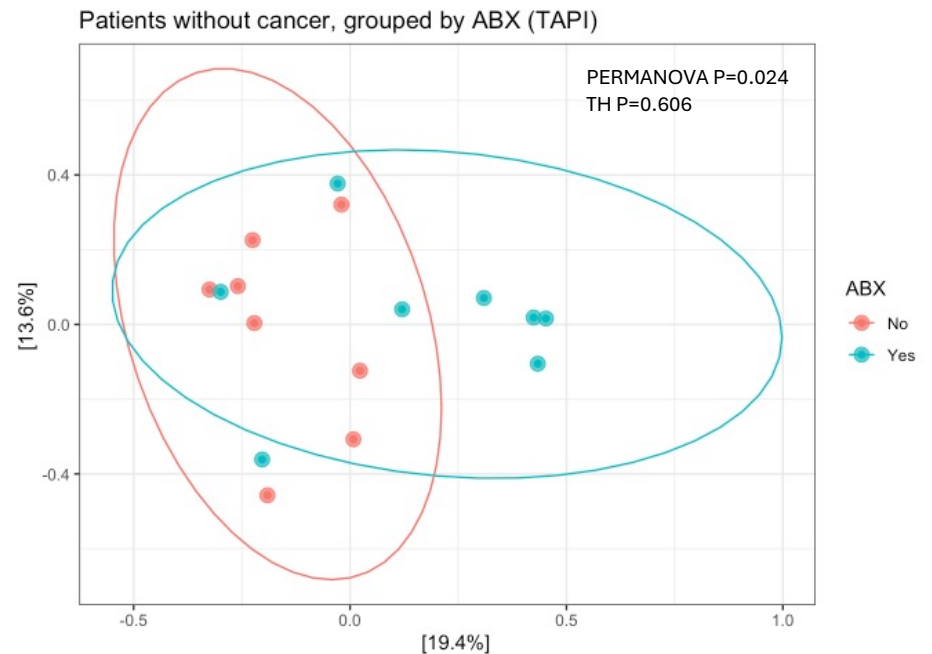
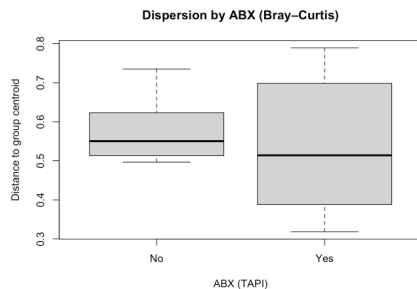
Adding metadata to microbiome associations: Constrained ordination

- Beta diversity
- Perform ordination using PCoA
 - Dissimilarity calculated using Bray-Curtis



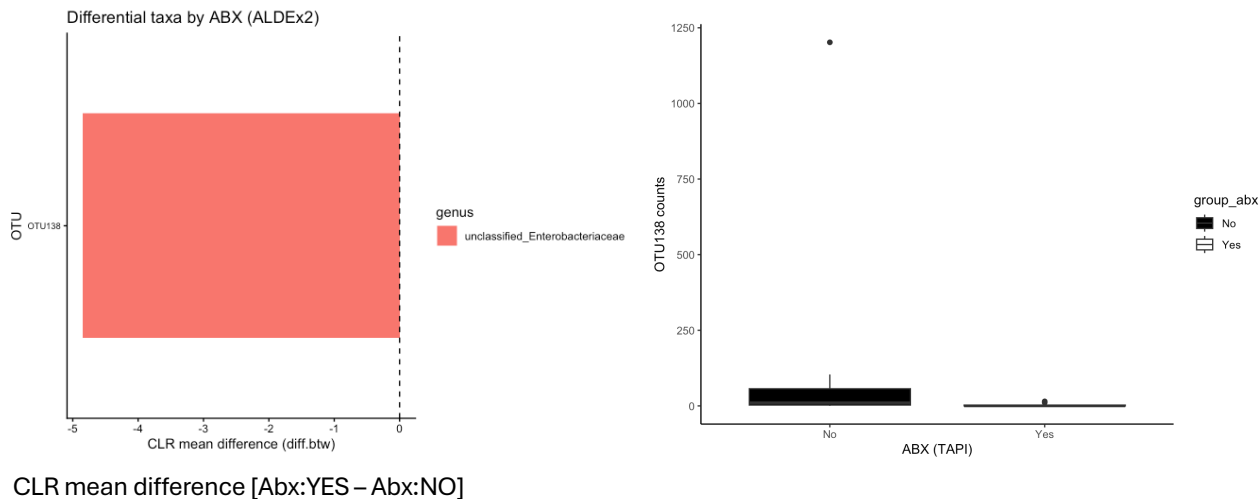
Adding metadata to microbiome associations: Constrained ordination

- **Are the group centroids different in multivariate space?**
 - PERMANOVA
- **Do groups have similar within-group spread?**
 - Beta dispersion or Test of homogeneity (TH)
- PERMANOVA p-value is small ($P < 0.05$)
 - Composition differs by ABX (TAPI)
- Beta dispersion is not significant



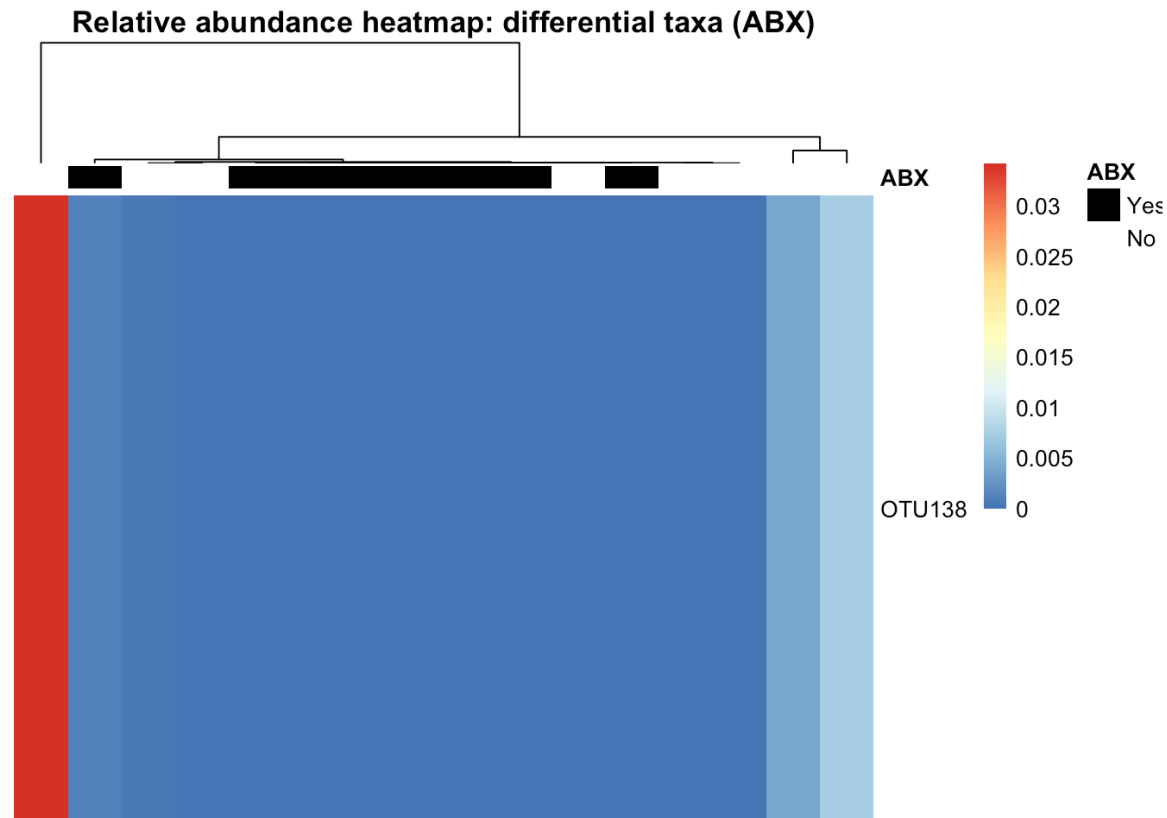
Adding metadata to microbiome associations: Differential Analysis

- **Which taxa are changing between ABX groups?**
- Microbiome data is compositional, ALDEx2 is most appropriate
- ALDEx2 method:
 - Performs CLR transformation
 - Models sampling uncertainty
 - Reports both statistical significance and effect size



Adding metadata to microbiome associations: Correlation heatmaps

- Using pheatmap we can create heatmap that illustrates the differences in relative abundance by ABX



Conclusion

- You should now be able to:
 - ✓ Understand the structure of microbiome sequencing data
 - ✓ Work with phyloseq objects for microbiome analysis
 - ✓ Visualize microbial composition within and between samples
 - ✓ Interpret alpha and beta diversity metrics
 - ✓ Explore taxa driving community differences
 - ✓ Build and interpret co-occurrence networks responsibly
 - ✓ Link microbiome patterns to clinical metadata
 - ✓ Generate publication-quality figures using reproducible code

Final Remarks

THANK YOU to GCC staff and the AMR26 conference organizers